Organic Chemistry

Chapter 2 — Hydrocarbons, Functional Groups, and Geometry

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2.1 Methane and Tetrahedral Geometry

Methane (CH_4) is the simplest hydrocarbon and the prototypical example of sp^3 hybridization. The carbon atom forms four equivalent sigma bonds to hydrogen atoms arranged at the vertices of a regular tetrahedron, with H–C–H bond angles of approximately 109.5 degrees and C–H bond lengths near 109 picometers. This geometry minimizes electron-pair repulsion in accordance with VSEPR theory.

Methane is the principal component of natural gas and a potent greenhouse gas, with a global warming potential roughly 28 times that of carbon dioxide over a century. Its combustion follows the equation $CH_4 + 2\ O_2$!’ $CO_2 + 2\ H_2O$, releasing approximately 890 kJ per mole. Visualizing the three-dimensional tetrahedral arrangement is essential to understanding why methane is non-polar despite the polarity of individual C–H bonds.

2.2 The Water Molecule and Hydrogen Bonding

Water (H_2O) consists of an oxygen atom bonded to two hydrogen atoms with an H–O–H bond angle of about 104.5 degrees, slightly less than the ideal tetrahedral angle because of repulsion from the two lone pairs on oxygen. The molecule is bent and strongly polar, with a dipole moment of 1.85 debye pointing from the hydrogens toward the oxygen.

Each water molecule can participate in up to four hydrogen bonds — two as donor through its hydrogens, two as acceptor through the oxygen lone pairs. This network underlies water's remarkable properties: high boiling point, high specific heat, surface tension, and the lower density of ice relative to liquid water. Visualization of the lattice of hydrogen-bonded water molecules in ice reveals a hexagonal open structure that explains why ice floats.

2.3 Benzene and Aromaticity

Benzene (C_6H_6) is a planar six-membered ring of carbon atoms, each bonded to a single hydrogen. The C–C bond lengths are all 139 picometers — intermediate between a single bond (154 pm) and a double bond (134 pm) — reflecting the delocalization of six pi electrons across the ring. This continuous pi system gives benzene its aromatic stability.

Hückel's rule states that a planar, fully conjugated monocyclic system is aromatic if it contains $4n+2$ pi electrons. Benzene satisfies the rule with $n = 1$, accounting for its enhanced stability and characteristic chemistry: it preferentially undergoes electrophilic aromatic substitution rather than addition. Visualizing the molecular orbitals — three bonding and three antibonding pi MOs distributed above and below the ring plane — clarifies why aromaticity is fundamentally a quantum-mechanical phenomenon.